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# Bihar Engineering University, Patna

## B.Tech 1st Semester Exam., 2021

***Subject: Mathematics—I (Calculus and Linear Algebra)***

***Code: 102102***

***Time: 03 Hours***

***Full Marks: 70***

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are NINE questions in this paper.*
*(iii) Attempt FIVE questions in all.*
*(iv) Question No. 1 is compulsory.*

***Q.1 Choose the correct answer (any seven) :*** **[2×7=14]**

(a) If $Y = \int_0^a t^4 e^{-2t^2} dt$, then the value of $Y$ is

(i) $[\sqrt{(2\pi)}] / [64]$ (iii) $[5\sqrt{(2\pi)}] / [64]$
(ii) $[3\sqrt{(\pi)}] / [64]$ (iv) $[3\sqrt{(2\pi)}] / [64]$

(b) The maximum area of a sector whose perimeter is $k$ is given by

(i) $k^2/16$ (iii) $k^2/4$
(ii) $k^2/8$ (iv) $k^2/1$

(c) The function $f(x) = x^2$, & $0 \le x \le 1/2$
$x$, & $1/2 < x \le 1$, then $\lim_{x \to 1/2} f(x)$

(i) has the value 1/2 (iii) has the value 2
(ii) has the value 1 (iv) does not exist

(d) If α,β are the roots of the equation $ax^2 + bx + c = 0$, then $\lim_{x \to \alpha} [1 - \cos(\alpha x^2 + bx + c)] / [(x - \alpha)^2]$ is equal to

(i) $a^2/2(\alpha-\beta)^2$ (iii) $a^2/2(\alpha-\beta)^2$
(ii) $-a^2/2(\alpha-\beta)^2$ (iv) 0

(e) The series $x - x^2/2 + x^3/3 - x^4/4 + \ldots$ converges for

(i) $(-1 < x \le 1)$ (iii) $|x| \le 1$
(ii) $|x| < 1$ only (iv) all real values of $x$

(f) If $u_n = \sqrt{(n+1)} - \sqrt{(n)}, v_n = \sqrt{(n^4 + 1)} - n^2$, then

(i) $\sum_{n=1}^{\infty} u_n$ converges but $\sum_{n=1}^{\infty} v_n$ diverges (iii) $\sum_{n=1}^{\infty} u_n$ and $\sum_{n=1}^{\infty} v_n$ both converge
(ii) $\sum_{n=1}^{\infty} u_n$ diverges but $\sum_{n=1}^{\infty} v_n$ converges (iv) $\sum_{n=1}^{\infty} u_n$ and $\sum_{n=1}^{\infty} v_n$ both diverge

(g) $\lim_{(x, y) \to (0, 0)} [1 - x - y] / [x^2 + y^2]$ is

(i) 0 (iii) -1
(ii) 1 (iv) Does not exist

(h) The gradient of the function $f(x, y, z) = \log(x^2 + y^2 + z^2)$ at $(3, -4, 5)$ is

(i) $1/\sqrt{(3)}(i + j + k)$ (iii) $1/25(3i^2 - 4j + 5k)$
(ii) $1/3(i + j + k)$ (iv) $1/25(3i^2 + 4j + 5k)$

(i) If A and $B$ are any two square matrices of order 2×2, then $(A + B)^2$ is equal to

(i) $A^2 + 2AB + B^2$ (iii) $A^2 + AB + BA + B^2$
(ii) $A^2 + AB - BA + B^2$ (iv) $A^2 + 2BA + B^2$

(j) The system of equations $x - y + 3z = 4$, $x + z = 2$, $x + y - z = 0$ has

(i) infinitely many solutions (iii) a unique solution
(ii) finitely many solutions (iv) no solution

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***Q.2 Solve both questions :*** **[10+4=14]**

(a) Evaluate $\int_0^\infty x^m e^{-ax^n} dx$.

(b) Find the surface of the solid formed by the revolution about the axis of $y$, of the part of the curve $ay^2 = x^3$ from $x = 0$ to $x = 4a$, which is above the $x$-axis.

***Q.3 Solve both questions :*** **[10+4=14]**

(a) Verify the Rolle's theorem for the function $f(x) = 2x^3 + x^2 - 4x - 2$.

(b) Find $\lim_{x \to \pi/2} (\sin x)^{\tan x}$.

***Q.4 Solve both questions :*** **[7+7=14]**

(a) Discuss the convergence of the sequence whose $n$ th term is $a_n = 1/\log n$.

(b) Test the convergence of the following series : $1 + x/1 + 1/2 \cdot x^3/3 + [1 \cdot 3 \cdot x^5] / [2 \cdot 4 \cdot 5] + [1 \cdot 3 \cdot 5 \cdot x^7] / [2 \cdot 4 \cdot 5 \cdot 6 \cdot 7] + cdots$

***Q.5 Solve both questions :*** **[7+7=14]**

(a) Find the Fourier series expansion of the function $f(x) = \{\pi + x\}$, $-\pi \le x \le \pi$. Hence deduce that $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + cdots$

(b) Find the Fourier cosine series and Fourier sine series of the following function in given interval : $f(x) = \{x + x^2\}$, $0 < x < 1$

***Q.6 Solve both questions :*** **[7+7=14]**

(a) Discuss continuity of the following function at the point $(-1, c)$ : $f(x, y) = [x^2 y] / [(1 + x)]$, & $x \neq -1$ $y$, & $(x, y) = (-1, c)$

(b) Find the extreme value of $xyz$, when $x + y + z = a$, $a > 0$.

***Q.7 Solve both questions :*** **[7+7=14]**

(a) Find the maximum value of the function $f(x) = 1/x$.

(b) Find the equation of the tangent plane to the surface $xy + yz + zx = -1$, at the point $(1, -1, 2)$.

***Q.8 Solve this question :*** **[14]**

Find the eigenvalues and eigenvectors of the following matrix : $\begin{vmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{vmatrix}$

***Q.9 Solve both questions :*** **[7+7=14]**

(a) Verify Cayley-Hamilton theorem for the matrix $\begin{vmatrix} 0 & c & b \\ -c & 0 & a \\ b & -a & 0 \end{vmatrix}$

(b) Determine the range of the following linear transformation. Also find the rank of $T$, where it exists, $T: V_3 \to V_3$, defined by $T(x_1, x_2, x_3) = (1/2x_1 + x_2 + x_3, x_1 - 1/3x_2, x_3)$

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